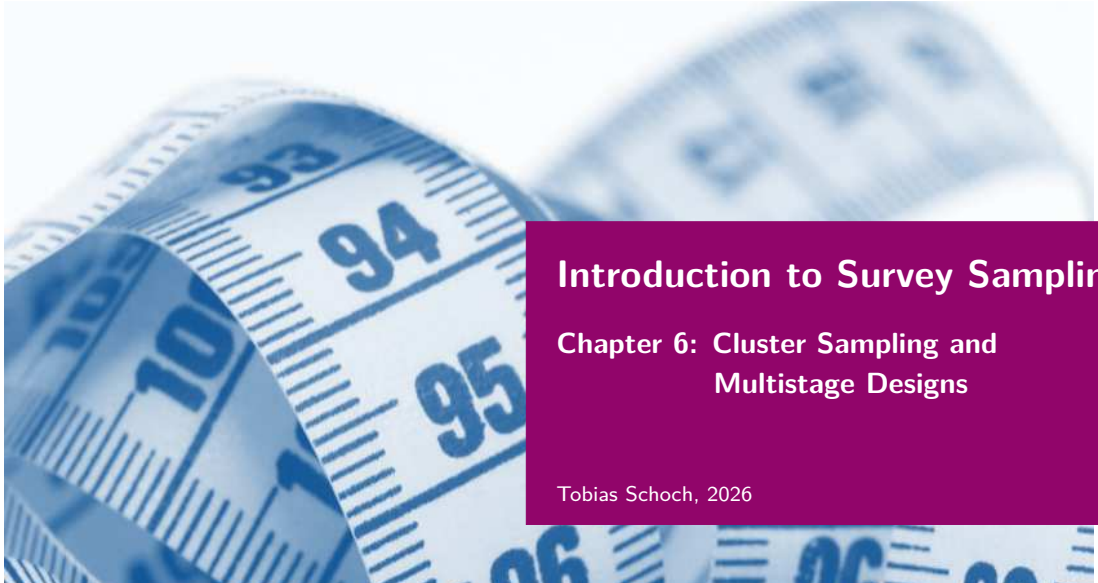
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Introduction to Survey Sampling

Chapter 6: Cluster Sampling and Multistage Designs

Tobias Schoch, 2026

Overview

6 Cluster sampling and multistage designs (brief overview only)

6.1 One-stage cluster sampling

6.2 Two-stage and multistage designs

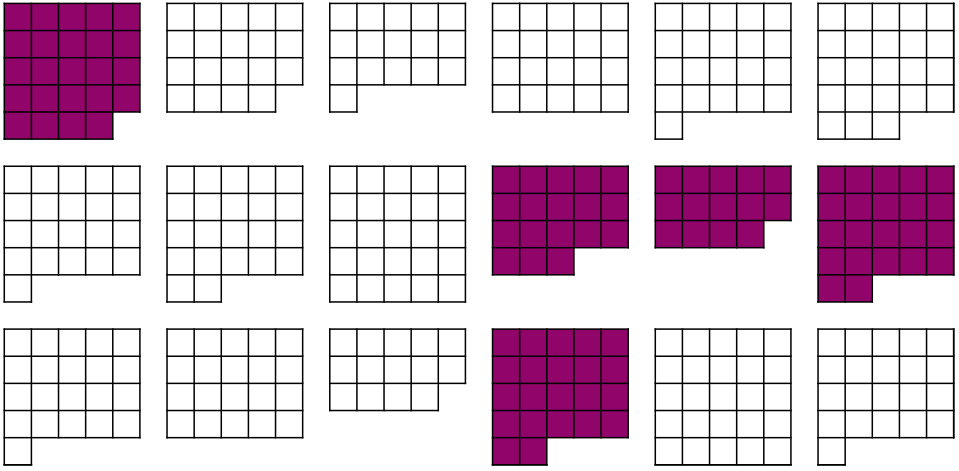


Figure: Cluster sampling: SRS of $n = 5$ from $N = 18$ school classes

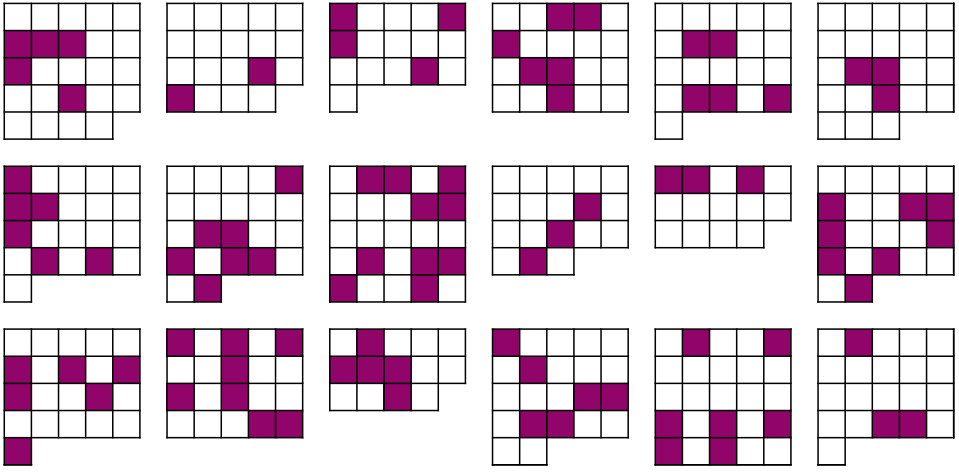


Figure: Simple random sample of $n = 100$ students

6 Cluster sampling and multistage designs

- **Population** of school classes: $U = \{1, \ldots, 18\}$ (see example)
- **Goal**: estimate the population total t_y of **project hours** (y_i) per week (focus: population; analyses at other levels are also possible)
- **Procedure**:
  - Select an SRS of $n = 5$ **clusters** (school classes) from U.
  - Observe **all elements** (students) in the selected clusters (school classes).
- **Reasons** for using this approach:
  - cost savings and economies of scale
  - the cluster total may itself be of interest, e.g., total income of all persons in a household

  - constructing a sampling frame of elements may be impossible or too expensive (e.g., all customers in a store on a Friday afternoon or the number of wild boars in a region)

- **Clustering** and **stratification** are similar in that both involve **grouping** population elements. The sampling procedure, however, is different.
- Whereas stratification can **increase precision** relative to SRS, clustering generally **reduces precision**.

The notation is somewhat **cumbersome**.

$N =$ number of clusters in U, — number of school classes

$M_i =$ number of elements in cluster i $(i = 1, \ldots, N)$, — number of students in class i

$M_0 = \sum_{i=1}^{N} M_i =$ total number of elements, — total number of students

$t_{yi} = \sum_{j=1}^{M_i} y_{ij} =$ total in cluster i, — total project hours in class i

$t_y = \sum_{i=1}^{N} t_{yi} = \sum_{i=1}^{N} \sum_{j=1}^{M_i} y_{ij} =$ population total, — total project hours

$$\sigma_T^2 = \frac{1}{N-1}\sum_{i=1}^{N}\left(t_{yi} - \frac{t}{N}\right)^2 = \text{var. of the cluster tot.},$$

$$\bar{y}_U = \sum_{i=1}^{N}\sum_{j=1}^{M_i}\frac{y_{ij}}{M_0} = \text{population mean},$$

mean number of project hours per student

$$\bar{y}_{i,U} = \sum_{j=1}^{M_i}\frac{y_{ij}}{M_i} = \frac{t_{yi}}{M_i} = \text{mean in cluster } i,$$

mean number of project hours per student in class i

6.1 One-stage cluster sampling

1) Select an SRS s of $n < N$ **clusters** (school classes) from U.
2) **Observe all** M_i elements (students) in each cluster selected in Step 1, $i \in s$.

Estimating the population total

An unbiased **estimator** of t_y is

$$\widehat{t}_y = N\,\bar{t}_y, \qquad \text{with} \qquad \bar{t}_y = \frac{1}{n}\sum_{i\in s} t_{yi},$$

with theoretical **variance**

$$\text{var}_p(\widehat{t}_y) = N^2\left(1-\frac{n}{N}\right)\frac{\sigma_T^2}{n}.$$

An unbiased **variance estimator** is

$$\widehat{\mathrm{var}}_p(\widehat{t}_y) = N^2 \left(1 - \frac{n}{N}\right) \frac{\widehat{\sigma}_T^2}{n} \qquad \text{with} \qquad \widehat{\sigma}_T^2 = \frac{1}{n-1} \sum_{i \in s} (t_{yi} - \bar{t}_y)^2 .$$

Estimating the population mean. We distinguish **two cases**:

1) The total number of elements M_0 is **known**. The population mean $\bar{y}_U$ can be estimated unbiasedly by

$$\bar{y}_{\mathrm{unv}} = \frac{\widehat{t}_y}{M_0}, \qquad \text{with} \quad M_0 = \sum_{i=1}^{N} M_i,$$

with **variance estimator** $\widehat{\mathrm{var}}_p(\bar{y}_{\mathrm{unv}}) = \dfrac{\widehat{\mathrm{var}}_p(\widehat{t}_y)}{M_0^2}$.

2) The total number of elements M_0 is **unknown**. The population mean can be estimated by

$$\bar{y}_{\text{rat}} = \frac{\sum_{i \in s} t_{yi}}{\sum_{i \in s} M_i} = \frac{\bar{t}_y}{\overline{M}},$$

where $\overline{M}$ is the sample mean of the cluster sizes M_i, $i \in s$,

$$\overline{M} = \frac{1}{n} \sum_{i \in s} M_i.$$

The estimator $\bar{y}_{\text{rat}}$ is a **ratio estimator** and is **not unbiased** for $\bar{y}_U$. An approximate **variance estimator** is

$$\widehat{\text{var}}_p(\bar{y}_{\text{rat}}) \approx (1-f) \frac{1}{n \overline{M}^2} \sum_{i \in s} \frac{\left(t_{yi} - M_i \, \bar{y}_{\text{rat}}\right)^2}{n-1}.$$

Efficiency considerations: Design effect

- Recall the **design effect** (deff) from Chapter 2.
- Let U consist of N clusters of **equal size** M. For one-stage cluster sampling,

$$\text{deff} \approx 1 + (M-1)\rho,$$

  where ρ is the **intracluster correlation** (ICC).
- Typically, $\rho > 0$, so that clustering results in **deff $>$ 1** and therefore a loss in precision relative to SRS.
- If $\rho = 0$, then deff ≈ 1: clustering produces approximately the same precision as SRS.

6.2 Two-stage and multistage designs

- Two-stage or multistage designs select **subsamples** within clusters. A distinction is made between *multistage* **element** and *multistage* **cluster** *designs*. For a **two-stage element design**, the procedure is as follows:

  1) Select an SRS s of n **PSUs** (primary sampling units; here, school classes) from U.

  2) Select an SRS of **SSUs** (secondary sampling units; here, students) within each of the n PSUs $i \in s$. Denote the SRS of m_i SSUs from the M_i elements by s_i ($i \in s$).

- The **estimators** of population totals and means are **not much more complicated** than under one-stage cluster sampling; see, e.g., Wu and Thompson (2020, Chapter 3.4) or Lohr (2019, Chapter 5.3).

- The theoretical **variances**, however, are more **complicated** because sampling occurs in two stages.
- **Example:** Within PSU i, an unbiased estimator of the PSU total t_{yi} is

$$\widehat{t}_{yi} = \frac{M_i}{m_i} \sum_{j \in s_i} y_{ij}.$$

- An unbiased estimator of the population total t_y is therefore

$$\widehat{t}_y = \frac{N}{n} \sum_{i \in s} \widehat{t}_{yi}.$$

- Equivalently,

$$\widehat{t}_y = \sum_{i \in s} \sum_{j \in s_i} w_{ij} y_{ij}, \qquad w_{ij} = \frac{N M_i}{n m_i} = \frac{1}{(n/N)(m_i/M_i)}.$$

- The **theoretical** variance is

$$\text{var}_p(\widehat{t}_y) = \underbrace{N^2\left(1-\frac{n}{N}\right)\frac{\sigma_T^2}{n}}_{\text{Stage 1}} + \underbrace{\frac{N}{n}\sum_{i=1}^{N}\left(1-\frac{m_i}{M_i}\right)M_i^2\frac{\sigma_{yi}^2}{m_i}}_{\text{Stage 2}},$$

  where σ_{yi}^2 is the population variance of y within PSU i.

- **Stage 1:** uncertainty from selecting the PSUs and equals the variance under one-stage cluster sampling.

- **Stage 2:** additional uncertainty from subsampling elements within the selected PSUs.

  $\Rightarrow$ It is small when the within-PSU sampling fractions m_i/M_i are large and/or the within-PSU variances σ_{yi}^2 are small. If all elements in each selected PSU are sampled ($m_i = M_i$), the Stage 2 component is zero.

- The variance estimator follows the same **two-component structure** as the theoretical variance:

$$\widehat{\text{var}}_p(\widehat{t}_y) = \widehat{V}_{\text{Stage 1}} + \widehat{V}_{\text{Stage 2}}.$$

- The **Stage 1** component is estimated from the variability of the estimated PSU totals $\widehat{t}_{yi}$ across the sampled PSUs

  $\Rightarrow$ It uses the estimated PSU totals $\widehat{t}_{yi}$, not the unknown true totals t_{yi}.

- The **Stage 2** component is estimated from the within-PSU sample variances $\widehat{\sigma}^2_{yi}$ in the sampled PSUs.

Multistage and multiphase designs

- **Multistage sampling:** sampling proceeds through a **hierarchy** of sampling units. For example, first sample schools (PSUs), then students (SSUs) within the sampled schools.

  $\Rightarrow$ Second-stage samples are typically selected independently across PSUs, which simplifies variance calculations.

- **Multiphase sampling:** essentially the same type of units is sampled in successive phases, typically because different information is collected at different phases. For example, draw a large first-phase sample and collect inexpensive auxiliary variables, then select a subsample for more costly measurements.

References

Lohr, S. L. (2019). *Sampling: Design and Analysis*, Boca Raton (FL): Chapman and Hall/CRC, 2nd edition.

Wu, C. and Thompson, M. E. (2020). *Sampling Theory and Practice*, Cham: Springer.

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